Why Math for AI

Three examples, one destination, and the course map

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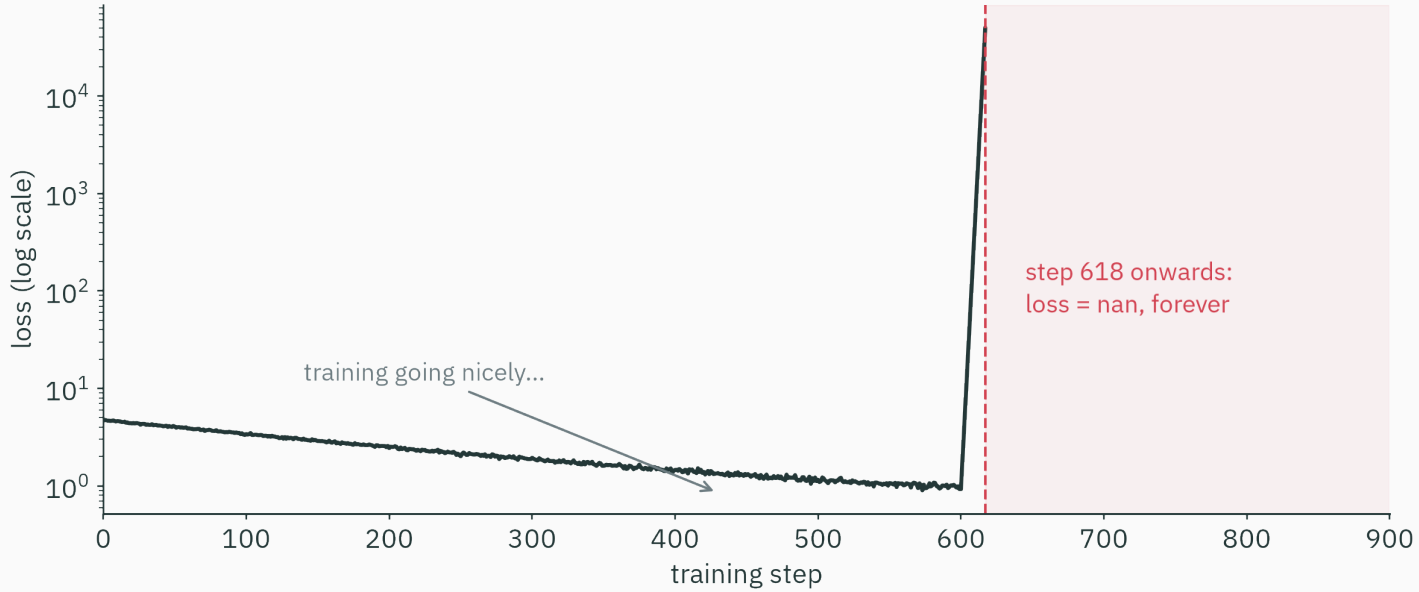
Mathematical Foundations for AI · IIT Gandhinagar

Three motivating examples

Example 1: a softmax loss becomes nan

V VISUAL

You train a model. The loss falls beautifully for 617 steps. Then —



No error message. No crash. Just nan, forever. **What happened?**

Reproduce the numerical failure in three lines

The model's last layer computes a **softmax** — it turns scores into probabilities:

```
>>> import numpy as np
>>> z = np.array([1000., 1001., 1002.])   # perfectly reasonable scores
>>> np.exp(z) / np.exp(z).sum()
array([nan, nan, nan])
```

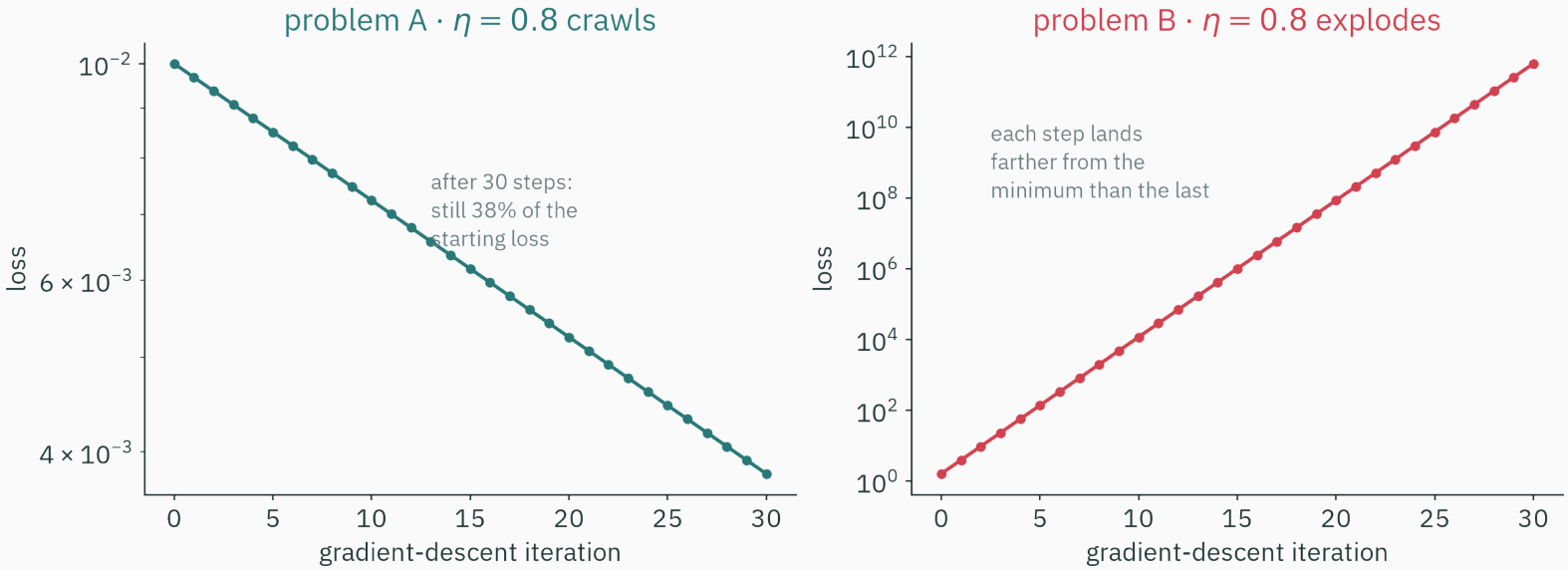
e^{1000} is larger than **any number this machine can store** — so it gives up.

Real numbers and machine numbers are **not the same thing**. The gap between them is **Lecture 2** — the very next class.

Example 2: one learning rate, two outcomes

V VISUAL

Gradient-based methods train many machine-learning models. Same update rule, same learning rate $\eta = 0.8$ — on two different problems:



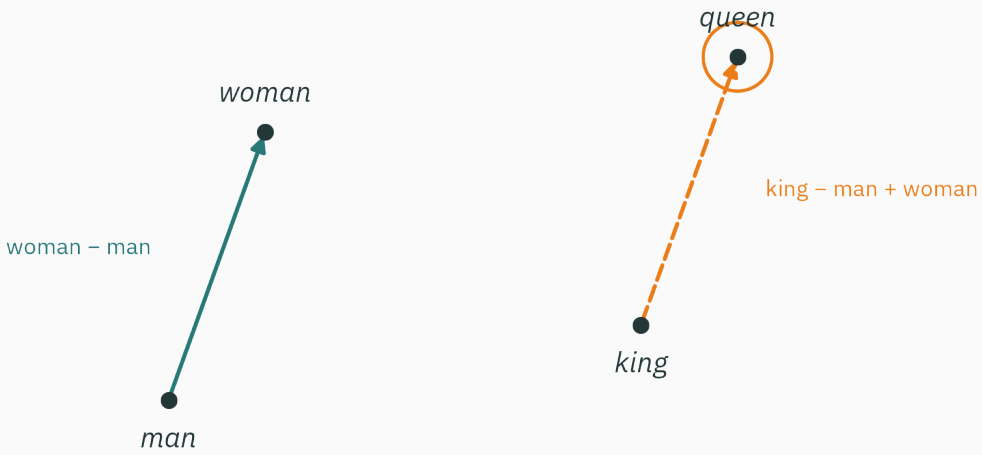
It crawls on one and explodes on the other. **Why?** Resolved in **L17** (conditioning).

Example 3: arithmetic with word embeddings

V VISUAL

A word-embedding model stores each vocabulary item as a vector of learned numbers. In one such space, this approximate relation appears:

$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$



a 2-D shadow of a 300-dimensional embedding space (→ L3)

Example 3: arithmetic with word embeddings

V VISUAL

How can **subtracting lists of numbers** capture *gender*? Resolved in **L3** (vectors).

Three examples, three mathematical explanations

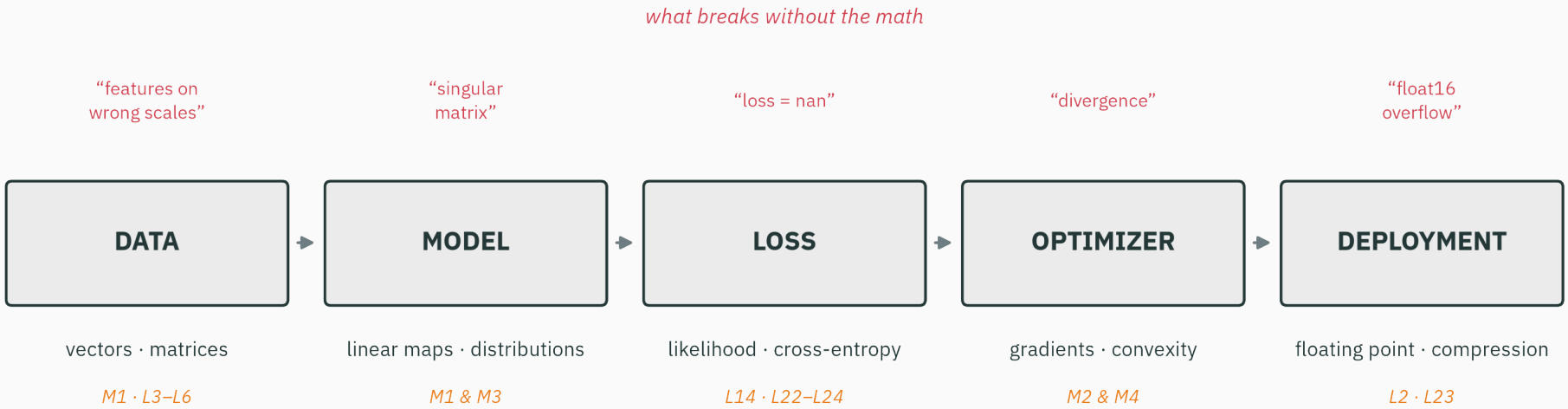
Example	The mathematics used to explain it	Lecture
the loss goes nan	floating-point numbers	**L2** — next class!
the same η crawls *and* explodes	conditioning, curvature	**L17**
king – man + woman ≈ queen	vector geometry	**L3**

Each example points to a mathematical question that the course will answer. The mathematics is the explanation, not decoration.

The AI stack

A useful five-stage view of an AI system

V VISUAL



the math each stage runs on — and the lectures that deliver it

The stack, stage by stage

Stage	The math it runs on	Where we build it
Data	vectors, matrices, norms	L3–L6 (stored as floats: L2)
Model	linear maps, probability distributions	L4–L6, L12–L13, L25
Loss	likelihood, cross-entropy	L14, L22–L24
Optimizer	gradients, autodiff, convexity	L7–L11, L16–L21
Deployment	floating point, compression	L2, L23

This table is the course: five stages, six modules, one destination.

The destination

By Lecture 26, you will train this

A **character-level language model**, from scratch, on the complete works of Shakespeare. Its actual output:

```
DUKE VINCENTIO:
Well, your wit is in the care of side and that.

CLARENCE:
And so the sight the world and the more,
I shall be my lord the king of hearth.
```

This is much smaller than GPT. What the two share is precise: both assign probabilities to text, fit parameters from data, use a cross-entropy objective, and train with gradient-based computation in floating point.

What each module contributes

What that little language model needs from each module:

Module	What the language model needs from it
0 · Machine numbers	probabilities underflow → log-space, log-sum-exp
1 · Linear algebra	transition matrices, stationary distributions
2 · Calculus + autodiff	gradients of the loss, backprop
3 · Probability + MLE	the model *is* a distribution; fitting = MLE
4 · Optimization	gradient descent actually finds the parameters
5 · Information theory	cross-entropy loss, perplexity
6 · Markov chains	the model class itself

We check the destination at every module boundary

After module…	…you can build this much of it
0 · machine numbers	store its probabilities without underflow
1 · linear algebra	run its transition matrix, find steady states
2 · calculus + autodiff	differentiate its loss automatically
3 · probability + MLE	define its loss (and know where losses come from)
4 · optimization	actually train it
5 · information theory	evaluate it (perplexity, compression)
6 · Markov chains	assemble the whole thing — and generate Shakespeare

At each module boundary, we can point to one more part of the final model that you can already explain or implement.

The teaching contract

How every idea in this course will arrive

1. **Picture** — a geometric intuition you can *see*
2. **Numbers** — a worked example small enough to check by hand
3. **Code** — the same example in ~5 lines of NumPy
4. **Symbols** — only *now*, the general definition

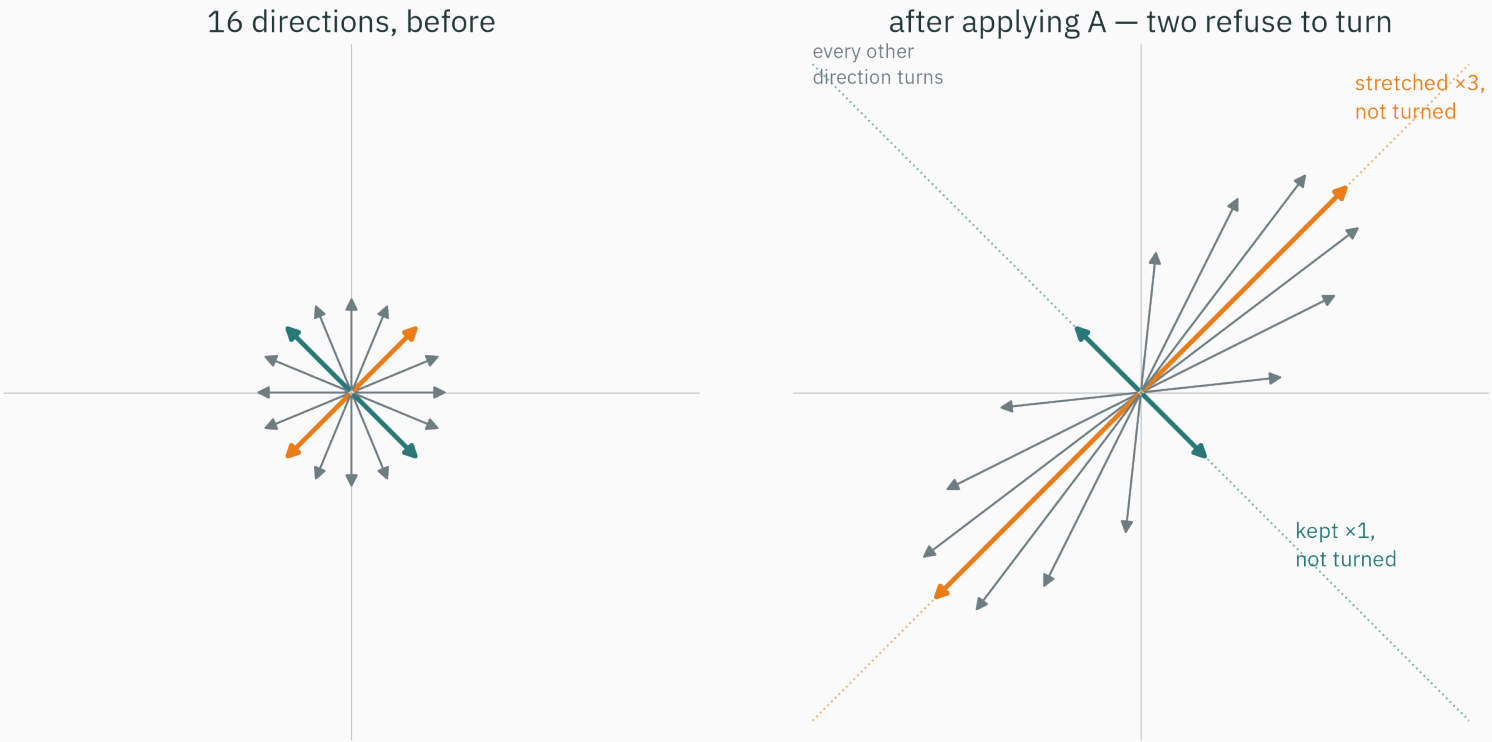
If you ever meet a formula in this course and think “where did that come from?” — that is a bug in the course. Report it.

Let’s demo the contract right now, on a concept from **L5**: *eigenvectors*.

Step 1 · picture: the directions that don’t turn

V VISUAL

A matrix moves every vector. Watch 16 directions, before and after:



Almost every direction gets **turned**. Two directions refuse.

Step 2 · numbers: check it by hand

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

The special direction $\boldsymbol{v} = (1, 1)$:

$$A\boldsymbol{v} = \begin{pmatrix} 2 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 2 \cdot 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3\boldsymbol{v}$$ same line, stretched $\times 3$

An ordinary direction $\boldsymbol{u} = (1, 0)$:

$$A\boldsymbol{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$ knocked off its line — turned

Step 3 · code: three lines of NumPy

```
>>> A = np.array([[2., 1.], [1., 2.]])
>>> vals, vecs = np.linalg.eig(A)
>>> vals
array([3., 1.])
>>> vecs[:, 0]                    # the (1,1) direction, normalized
array([0.70710678, 0.70710678])
```

The machine found both special directions — and how much each one stretches.

Step 4 · symbols: only now, the definition

A nonzero vector $\boldsymbol{v}$ is an **eigenvector** of A, with **eigenvalue** λ, if

$$A\boldsymbol{v} = \lambda\boldsymbol{v}$$

— applying A only *scales* $\boldsymbol{v}$; it never turns it.

You just previewed L5 — and seen how every concept will arrive.
Picture first. Symbols last. Always.

The fine print

Prerequisites · an honest accounting

We assume (ES 113 + ES 114)

- Python + NumPy — loops, arrays, plotting
- discrete probability — coins, dice, Bayes on events
- descriptive statistics — mean, variance
- school calculus — $\frac{\mathrm{d}}{\mathrm{d}x} x^n$, one variable

We do NOT assume

- any machine learning
- linear algebra beyond 12th grade
- multivariable calculus
- continuous distributions / densities
- optimization of any kind

If you can read a `for` loop and reason about a coin flip, you have everything you need on day one.

How the course runs

What	How much	Notes
Lectures	26 × 80 min	intuition first, with an AI example in every lecture
Tutorials	13 × 80 min	**hybrid**: ~40 min worksheet + ~40 min notebook
Credit	3–1–0–4	tutorials are formative — effort counts, not marks
Assessment	quizzes + midsem + endsem	a quiz after (nearly) every module

Worksheets build pen-and-paper fluency (exam style); notebooks rebuild the *same idea* computationally. You need both hands.

The textbooks · all excellent, all free

Book	Role in this course
Deisenroth, Faisal, Ong — *Mathematics for ML*	the backbone (Modules 1–4)
Boyd & Vandenberghe — *Convex Optimization*	Module 4 (optimization)
MacKay — *Information Theory, Inference & Learning*	Module 5 (information theory)
Solomon — *Numerical Algorithms*	Module 0 + numerics throughout

All four are **legally free PDFs** from the authors: mml-book.github.io · stanford.edu/~boyd/cvxbook · inference.org.uk/itila · Solomon's *Numerical Algorithms*. Reading pointers close every lecture.

The course site · everything in one place

- **Slides** — PDF for every lecture, posted before class
- **Worksheets** — pen-and-paper problem sets, with solutions
- **Notebooks** — every tutorial's computational half, Colab-ready
- **Interactives** — browser toys for the big ideas (eigenvectors, gradient descent, IEEE-754, …)

Tutorial 1 lands right after L2: IEEE-754 by hand (worksheet) + floating-point experiments in NumPy (notebook). Tutorials always follow the lecture that feeds them.

The map

The course at a glance

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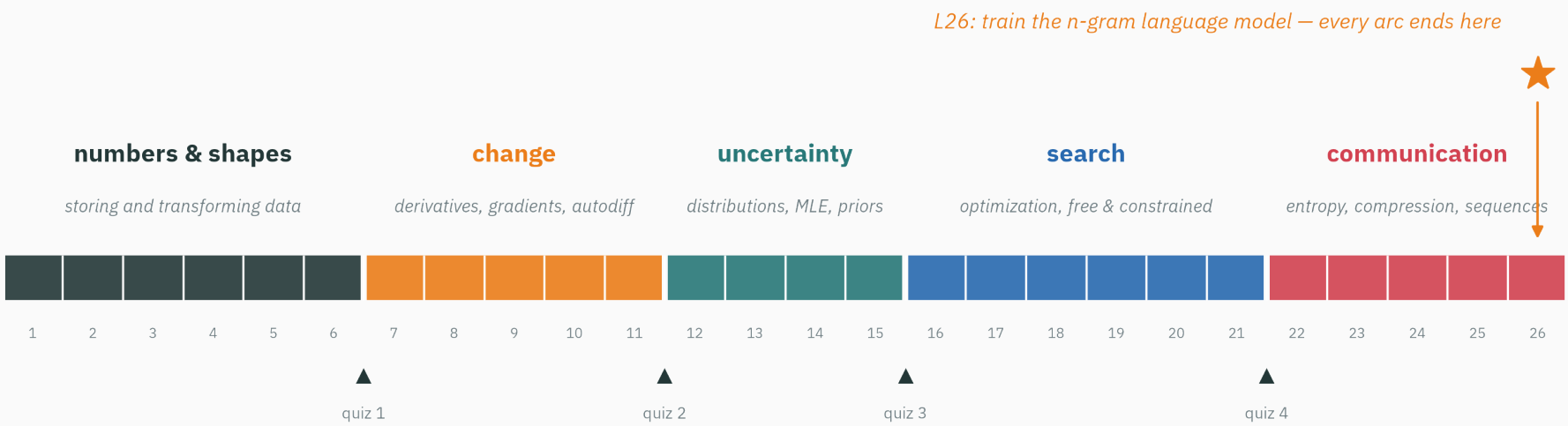

The course sequence

Lectures	Arc	The question it answers
L1–L6	**numbers & shapes**	how do machines store data — and transform it?
L7–L11	**change**	how do we measure sensitivity — and automate it?
L12–L15	**uncertainty**	how do we model data with distributions — and fit them?
L16–L21	**search**	how do we find the best parameters, free and constrained?
L22–L26	**communication**	entropy, compression, and sequence models

The course sequence

In L14, negative log-likelihood explains an important family of losses, including cross-entropy. Other objectives, such as hinge loss and explicit regularizers, have different origins.

Checkpoint: calibrate yourself

Q QUESTION

No marks — just checking that the prerequisites are alive:

Three warm-ups. Commit to answers for all three before the next slide.

A. You flip a fair coin twice. P(two heads)?

B. What does `np.dot([1, 2], [3, 4])` return?

C. What is the slope of $f(x) = x^2$ at $x = 3$?

D. (no fourth question — breathe)

Answers: calibrate yourself

A ANSWER

A. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ — independent events multiply (ES 114). By L12 you'll do the *continuous* version.

B. $1 \cdot 3 + 2 \cdot 4 = 11$ — the dot product. In L3 it becomes *geometry*: length, angle, similarity.

C. $f'(x) = 2x$, so 6 — school calculus. In L7 the derivative becomes *the best local linear model*.

If all three felt easy, you are fully prepared. If one wobbled, skim your ES 113/114 notes this week — these are the only prerequisites we lean on.

Lecture 1 — summary

- **AI fails (and works) in mathematically explainable ways** — NaN losses (L2), diverging optimizers (L17), embedding arithmetic (L3).
- **The stack**: data → model → loss → optimizer → deployment — each stage runs on specific math.
- **The destination**: a character-level language model, trained from scratch in L26.
- **The contract**: picture → numbers → code → symbols — demoed today on eigenvectors.
- **Prerequisites**: ES 113/114 only; no ML assumed. Four textbooks, all free.

Read before L2 — nothing required. If curious: Goldberg, *What Every Computer Scientist Should Know About Floating-Point Arithmetic* (1991), §1.

An AI system combines several mathematical components — by L26 you'll have built a small one from scratch.

Next: **floating point** — how machines store numbers, why e^{1000} overflows, and how stable softmax avoids it.